CSWO-Bench: A Multilingual Multimodal Benchmark of Citizen Science Water Observations

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Code and data: https://github.com/notEloiMm/cswo-bench

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1 Abstract

CSWO-Bench is a benchmark regarding citizen-science water observations. Here, data is sampled across six different platforms, harmonized and put into one singular schema. In total, this dataset contains 225,145 records. This report consists of the results of specifications 2–9 (corpus, statistics, extraction, gold set, agreement, baselines and image–text consistency), the bias audit of specification 11, and the literature and limitations of specification 14.

2 Introduction

Volunteers can report states of different water-bodies, and upload them to one of many citizen-science platforms. Each platform has its own schema, licence and language. We chose six of them: iNaturalist, CrowdWater [18], EyeOnWater, and three platforms published through GBIF [5]. We only kept open licences (CC0, CC-BY, ODbL, EyeOnWater's open policy) - each record carries its licence and provenance. The next sections cover the project specifications.

3 Spec 2 — Harmonised corpus

All sources were mapped into one schema containing note, English translation, photo URLs, date, coordinates rounded to 1 km, country, water-body type, labels, licence, provenance, hashed user id. The language was identified with lingua and notes were translated to English with OPUS-MT [19] and NLLB-200 [14], with an automatic check for empty, looping, truncated and hallucinated output. We masked personal names, e-mails and typed coordinates in notes.

Table 1: Records per platform and modality (%).

Platform	Records	Text only	Image only	Both	Main languages
iNaturalist	85,899	4.3	76.9	18.7	en
CrowdWater	65,892	0.4	78.0	21.5	en, de, et, es
Artportalen	28,223	100	0	0	sv
EyeOnWater	26,758	0	100	0	–
Laji.fi	16,707	100	0	0	fi
Other GBIF	1,670	100	0	0	da
Total	225,149				24 languages

Of the 80,886 notes, 32% are Swedish, 25% English, 16% Finnish and 14% undetermined. The other 20 languages are each under 5% (Table 1). Figure 1 shows the water-body type of the records per platform. Rivers and streams (36%) come almost all from CrowdWater, lakes and coasts from the Nordic portals and iNaturalist. Since EyeOnWater doesn't publish a locality type, a third is unknown.

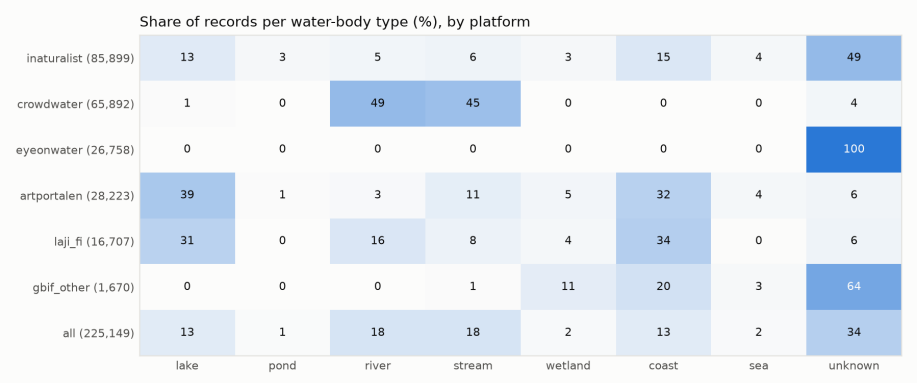


Figure 1: Share of records per water-body type and platform (%).

4 Spec 3 — Corpus statistics

Table 2 compares the notes with the Brown [4] and Reuters-21578 [11] corpora. The notes are rather short, and many words are only seen once. Their higher WordNet [13] coverage comes from missing function words. FinnWordNet [12] covers 72% of Finnish tokens.

Table 2: Field notes vs edited text.

	Field notes	Brown	Reuters
Mean length (tokens)	3.2 – 12.4	17.5	24.3
Median length	2 – 7	15	23
Words seen once	49 – 61%	43%	36%
Types in 25k tokens	4,622 – 5,910	6,327	4,448
Number tokens	5 – 15%	1%	10%
WordNet coverage (English)	76 – 82%	70%	74%

5 Spec 4 — Zipf and Heaps

Zipf [22]: OLS of log frequency on log rank, with the 90% confidence interval of the slope, R^2 and adjusted R^2 (Table 3, Figure 2). The law holds (R^2 0.95–0.98, like Brown and Reuters).

Table 3: Zipf fits by platform (spec 4): OLS of $\log_{10}$ frequency on $\log_{10}$ rank with 90% confidence interval of the slope.

platform	types	slope	90% lo	90% hi	R^2	adj. R^2	slope top-1000
inaturalist	16,209	-1.289	-1.292	-1.287	0.973	0.973	-0.952
artportalen	19,372	-1.157	-1.160	-1.154	0.954	0.954	-1.015
crowdwater	13,376	-1.087	-1.090	-1.084	0.963	0.963	-0.952
laji_fi	9,511	-0.972	-0.976	-0.968	0.950	0.950	-0.949
Brown	47,039	-1.323	-1.325	-1.321	0.974	0.974	-0.977
Reuters	29,188	-1.557	-1.560	-1.555	0.978	0.978	-0.961

Heaps [7]: vocabulary size V at every 5% of the tokens and is fitted as $V = KN^{\beta}$ (Table 4, Figure 3). Here, the field notes grow faster (β 0.55–0.68) than Brown (0.51) and Reuters (0.45).

Table 4: Heaps fits by platform (spec 4): $V = KN^{\beta}$ on vocabulary sizes at every 5% of the tokens.

platform	tokens	types	K	\beta	90% lo	90% hi	R^2
inaturalist	246,659	16,209	17.92	0.549	0.543	0.555	0.999
artportalen	224,669	19,372	8.71	0.626	0.623	0.629	1.000
crowdwater	103,315	13,376	10.37	0.622	0.616	0.628	1.000
laji_fi	52,767	9,511	7.78	0.654	0.651	0.658	1.000
Brown	1,006,060	47,039	44.82	0.505	0.494	0.516	0.997
Reuters	1,330,267	29,188	55.22	0.446	0.439	0.454	0.998

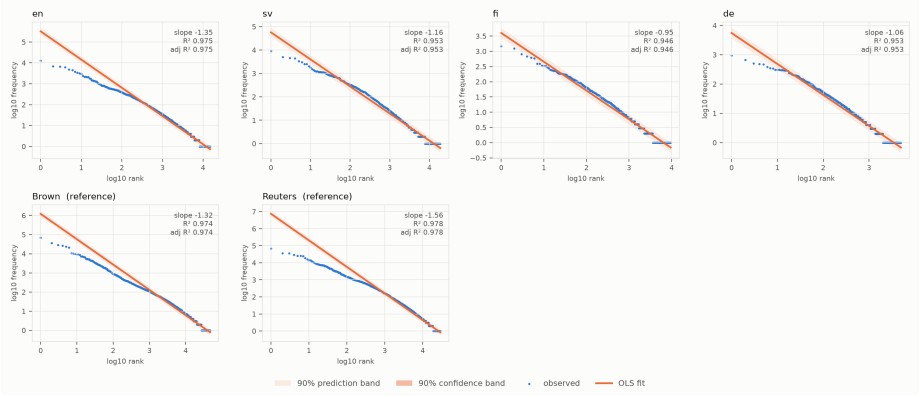


Figure 2: Zipf plots by language with OLS fit, 90% confidence band (dark) and 90% prediction band (light).

Table 5: New word types added in selected 5% steps of tokens, and the ratio of the last to the first step: growth slows down, much less for field notes than for edited text.

	5%	10%	25%	50%	75%	100%	last/first
inaturalist	3,065	1,520	870	596	442	415	0.14
artportalen	2,951	1,622	1,091	812	694	607	0.21
crowdwater	2,075	1,212	818	610	449	358	0.17
laji_fi	1,326	786	496	378	332	332	0.25
Brown	9,939	5,227	2,668	1,709	1,292	1,023	0.10
Reuters	7,508	3,143	1,621	952	750	548	0.07

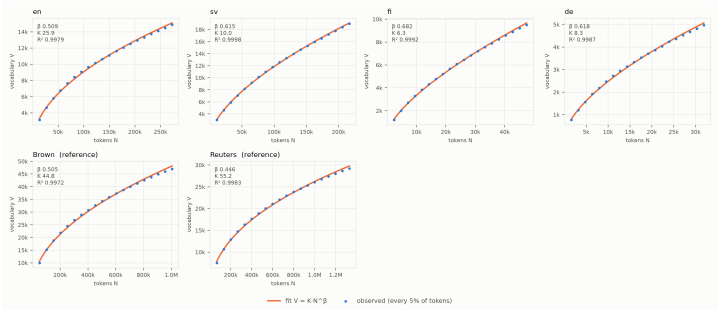


Figure 3: Heaps curves by language: vocabulary size against tokens, with fitted $V = KN^{\beta}$.

6 Spec 5 — Information extraction

We extract six entity types (water body, location, organism, quantity, date/time, weather) and twelve water conditions with three systems: spaCy pipelines for 18 languages [8], a multilingual XLM-RoBERTa NER model [3], and our rules (water-term lists in nine languages and regular expressions). The NER models know places, dates and numbers but not organisms, water bodies or weather, which only the rules cover. The same NER output was used to mask personal names: 8.8% of notes named a person, mostly survey credits.

7 Spec 6 — Gold set

We labelled 300 records from one stratified sample, plus 20 calibration records labelled by all three of us. Each gold record was labelled by one student and one LLM (Qwen2.5-VL-7B [1]) and a third person decided the disagreements. Table 6 summarises the labels.

Table 6: Gold set (300 records).

Label	Distribution
Relevance	about water 150, not 134, uncertain 16
Conditions	none stated 128; low or dry 16, high water 7, clear 6, turbid 4
Entities (144 spans)	water body 54, quantity 28, weather 18, location 16, organism 15, date 13
Image–text	consistent 94, cannot judge 11, inconsistent 5, partly 3

8 Spec 7 — Agreement and extractor evaluation

The agreement is measured with Cohen's κ [2] and span F1 (Table 7). The agreement between us students was high, but the agreement with the LLM is low - 68% of notes got marked as "not about water", while we students only labelled 40% this way.

Table 7: Agreement: relevance and condition κ, span F1 (exact / partial).

Pair	Relevance κ	Conditions κ	Span F1
Students, calibration (20)	0.80	1.00	1.00 / 1.00
Student vs LLM, gold (300)	0.25	0.29	0.20 / 0.29

The extractors are scored on the gold entity spans (Table 8). Exact scores are low because span boundaries differ from the guideline. With partial matching the rules are best. For water conditions the rules reach F1 0.31 (precision 0.50, recall 0.22). Only 51% of the gold notes are actually about water. Translating non-English notes to English helps condition extraction but hurts entity extraction, because translation garbles names and abbreviations (Table 9).

Table 8: Extractors on the gold set (150 spans, micro F1).

System	F1 exact	F1 partial	Predicted
spaCy	0.20	0.34	96
XLM-R NER	0.20	0.28	65
Rules	0.12	**0.44**	210
Union	**0.22**	0.40	332

Table 9: Extraction F1 on English originals vs non-English notes, original and machine-translated.

Text	spaCy entities	Rules entities	Rules conditions
English originals	0.42	0.53	0.39
Non-English, original	0.28	0.37	0.22
Non-English, translated	0.18	0.32	0.37

9 Spec 8 — Weak-label baselines

Using just the note, we can predict the Artportalen habitat field, as well as the CrowdWater water-level class. The dataset was split 80/10/10, stratified by platform; each CrowdWater gauge was kept in one split, which results in a 64/8/28 split for CrowdWater. Due to EyeOnWater having no notes, we cannot predict its Forel-Ule colour field from text. The models used were a TF-IDF with a linear SVM, and a multilingual DistilBERT [17], with and without stopwords (Table 10).

Table 10: Weak-label baselines, test macro F1 (117 habitat, 578 water-level notes).

Model	Habitat	Water level
Majority class	0.05	0.04
TF-IDF + SVM	**0.73**	**0.22**
DistilBERT	0.65	0.15

From the results, we can see that habitat is much easier to predict from the note than water level. This is thanks to the fact that habitat is often mentioned in the note; while water level is not, since it is read from a gauge in the photo. A surprising result is that the simple SVM had superior performance to DistilBERT, which is a much more complex model. Removing stopwords didn't affect much the performance of the models, with the macro f1 score changing by less than 0.01 in all cases.

10 Spec 9 — Image–text consistency

We compare each note against its photo with the cosine similarity of CLIP [16] (English text) and multilingual SigLIP [21] embeddings, on 148 gold records with a photo, as well as 600 other pairs. Due to only 8 gold records being inconsistent, we decided to score as well each note against a different, random record's photo, with the expectation being that a useful model should prefer the real pair (Table 11, Figure 4).

Table 11: Image–text similarity: AUC of own vs another photo, and against the gold labels.

Model	Own vs other	iNaturalist	CrowdWater	Gold (consistent vs not)
CLIP (English)	0.72	0.80	0.64	0.73
SigLIP (original)	0.70	0.75	0.66	0.70
SigLIP (translation)	0.69	0.75	0.63	0.71

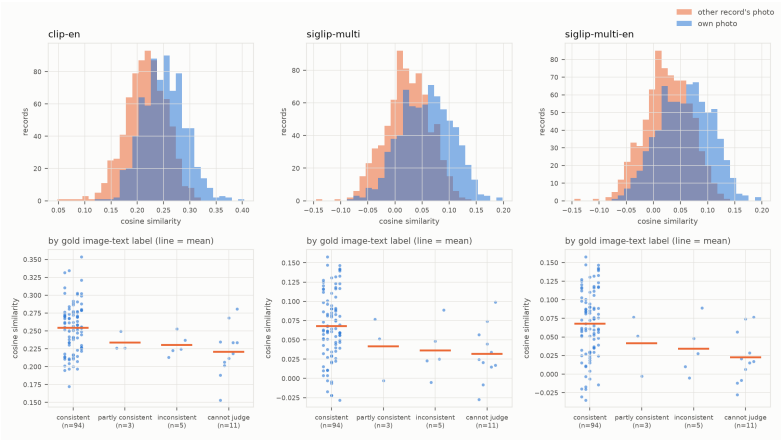


Figure 4: Similarity of a note with its own photo vs another record's photo (top) and by gold label (bottom).

From the results we can conclude that the real pair is preferred about 75% of the time, which is to say about 3 times in 4. This score is well above random chance, but it is not reliable. It works well on iNaturalist, where the note names the species in the photo, but quite poorly on CrowdWater where all photos show a riverbank with a gauge. This is due to the fact that the models measure whether photo and note are about the same topic, but not whether the photo shows the condition the note describes.

Table 12: Bias audit summary (Gini: 0 = equal, 1 = one unit has everything).

Dimension	Finding
Geography	148 countries; Sweden, US and Switzerland hold 37% of records (Gini 0.91); Africa, South Asia and South America nearly absent; Latvia only 264 records.
Language	Swedish, English and Finnish are 72% of the notes (Gini 0.82).
Season	Summer 37%, spring 29%, autumn 20%, winter 15%.
Users	The top 1% of users wrote 20–61% of a platform's records (Gini 0.65–0.88); CrowdWater has no user id.

11 Spec 11 — Bias audit

The corpus is significantly biased (Table 12, Figures 5 and 6). This is partly a result of our own collection choices, since iNaturalist was collected for 14 countries and three sources are national portals. Only CrowdWater and EyeOnWater are global. Records peak mostly in summer, meaning that winter states such as ice are rare (only two gold records). Additionally, on Laji.fi, about nine people (the top 1% of users) wrote 61% of the records, so in the Finnish text, the abbreviations they use dominate. This means that the results on CSWO-Bench should be reported per platform and language, and shouldn't be transferred to other regions, seasons or languages without previous analysis.



Figure 5: Where the records are: density of coordinates per platform (longitude × latitude).

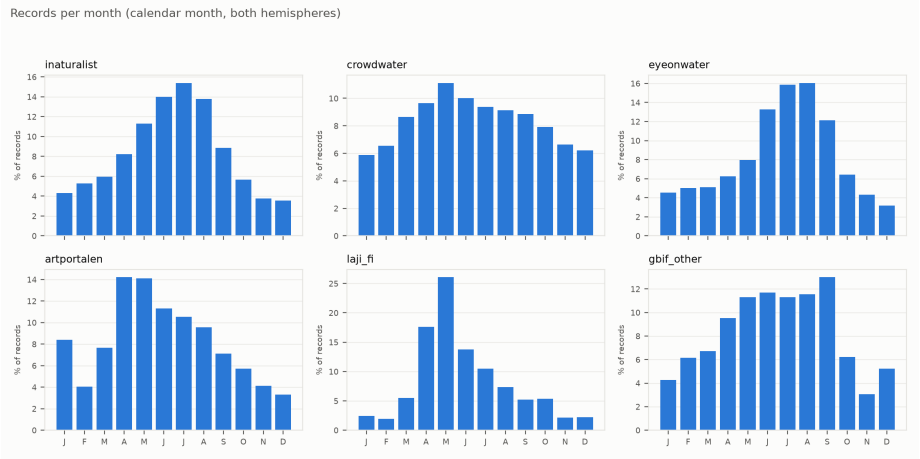


Figure 6: Share of records per calendar month, by platform.

12 Spec 14 — Literature and limitations

Literature. CrowdWater's virtual staff gauges [18] and GBIF's national portals [5] provide the structured data we build on; Kosmala et al. [10] show that citizen-science data can be reliable when quality is checked, which motivates our gold set and bias audit. Our statistics follow Zipf [22] and Heaps [7]; agreement uses Cohen's κ [2]; the data card follows Datasheets for Datasets [6]. Beyond specs 2–9, the repository also contains a zero-shot LLM/VLM evaluation (Qwen2.5 [15, 1]; 68–69% of checked outputs hallucinate, in the sense of [9]) and a GLiNER comparison [20] that raises partial entity F1 from 0.44 to 0.52.

Limitations.

- *Gold design:* one student and one LLM per record (course guideline), not two students; student–student agreement comes only from the 20 calibration records.
- *Split:* CrowdWater is 64/8/28 instead of 80/10/10, to keep each gauge in one split.
- *Small numbers:* few positive gold labels (5 inconsistent image–text pairs, 4 turbid notes).
- *Pipeline errors:* language identification fails on species names; translations are plausible but not always correct; the water filter is keyword-based (only 51% of gold notes are about water).
- *Coverage and privacy:* Latvian data is almost absent (non-commercial licences excluded); name masking misses a few names; photos are linked, not copied.

13 Code availability

All source code, the fixed splits, the gold labels, the results and the data card are in the repository https://github.com/notEloiMm/cswo-bench (MIT licence for the code). The pipeline is a Python package (`src/cswo/`) with one command per step listed in the README. The steps that need a GPU are Jupyter notebooks in `notebooks/` that run on a free Google Colab T4:

- `colab_translate.ipynb` — translation to English (spec 2);
- `colab_llm_annotate.ipynb` — LLM second annotator for the gold set (spec 6);
- `colab_bert.ipynb` — DistilBERT baselines (spec 8);
- `colab_clip.ipynb` — CLIP/SigLIP image–text similarity (spec 9);
- `colab_zeroshot.ipynb` — zero-shot LLM/VLM evaluation (spec 10).

The evaluation scripts read only committed files (gold labels, extractor outputs, model predictions), so every table of this report can be reproduced without the full corpus.

14 Use of AI tools

AI assistance in producing the project. We used LLMs in different context throughout the project. We used Claude for generating tedious coding tasks and autocompletion - always supervised. It was also used as help for plotting plots. Further we used it for generating scripts and the generation of latex tables - with Gemini.

References

[1] Shuai Bai et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[2] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

[3] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[4] W. Nelson Francis and Henry Kučera. *Brown Corpus Manual*. Department of Linguistics, Brown University, 1979.

[5] GBIF.org. GBIF: The global biodiversity information facility. https://www.gbif.org, 2026. Download DOIs per record source in `data/licence_log.csv`.

[6] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.

[7] Harold Stanley Heaps. *Information Retrieval: Computational and Theoretical Aspects*. Academic Press, 1978.

[8] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. spaCy: Industrial-strength natural language processing in Python. Zenodo, 2020.

[9] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023.

[10] Margaret Kosmala, Andrea Wiggins, Alexandra Swanson, and Brooke Simmons. Assessing data quality in citizen science. *Frontiers in Ecology and the Environment*, 14(10):551–560, 2016.

[11] David D. Lewis. Reuters-21578 text categorization test collection, distribution 1.0. AT&T Labs – Research, 1997.

[12] Krister Lindén and Lauri Carlson. FinnWordNet – WordNet på finska via översättning. *LexicoNordica*, 17, 2010.

[13] George A. Miller. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41, 1995.

[14] NLLB Team, Marta R. Costa-jussà, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.

[15] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

[17] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

[18] Jan Seibert, Barbara Strobl, Simon Etter, Patrick Hummer, and H. J. (Ilja) van Meerveld. Virtual staff gauges for crowd-based stream level observations. *Frontiers in Earth Science*, 7:70, 2019.

[19] Jörg Tiedemann and Santhosh Thottingal. OPUS-MT – building open translation services for the world. In *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation (EAMT)*, 2020.

[20] Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[21] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[22] George Kingsley Zipf. *Human Behavior and the Principle of Least Effort*. Addison-Wesley, 1949.